From Equity Pairs to Options Relative Value

Does learning improve capital allocation?

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Abstract

We extend equity-pairs trading to relative implied volatility in approximately 30-day, 25-delta puts on related stocks. A screened opportunity rule generates trades; learning then determines their entry size. In a simulated $10 million account over January 2022–August 2025, constant full sizing achieves annualized excess Sharpe 1.055, compared with 1.263 for decision-tree states and cross-entropy sizing. The observed gain of 0.208 has a conditional 95% interval of $[-0.103, 0.424]$. Learned sizing also changes which trades a constrained portfolio can admit. This motivates sequential Q-learning, but the tested policies do not establish an additional benefit. Concentrated profits, previously inspected history and simplified execution limit the conclusions. Learned sizing has a higher historical point estimate, but the comparison does not establish a repeatable advantage from learning.

1 Introduction

Suppose AMD and NVDA face similar news, but AMD options become unusually expensive relative to NVDA options. Selling the richer options and buying the cheaper ones expresses a view about the relative price of volatility. Daily stock hedges reduce immediate directional exposure, and initial vega matching reduces sensitivity to a common small change in implied volatility. The trade still carries risk: related stocks can experience different events, and option premiums may correctly reflect those differences.

Our starting point is Ning and Lee's equity-pairs framework [1], which separates constructing a spread from learning how to trade it. We carry that distinction into options. First, we ask whether a screened relative-volatility rule produces useful historical trades under explicit contract and funding assumptions. We then ask the central question: *does learning improve how capital is allocated to those opportunities?* A profitable signal and a useful allocation policy are separate findings.

After explaining the motivating paper and our adaptation, we define the trade and evaluation. We compare constant sizing with learned entry sizing, then examine why shared portfolio constraints affect the comparison. Finally, we test whether a policy that values future opportunities can improve on simpler decisions. All results come from existing saved experiments; this revision introduces no new backtest or model selection.

2 Motivating paper and adaptation

What the paper does

Ning and Lee [1] study how to construct and trade mean-reverting equity spreads. For a pair they form $X_t = S_{1,t} - BS_{2,t}$ and choose B by grid search to minimize empirical mean reversion time: the average historical duration from a significant local extreme to the next crossing of the sample mean. This forms a spread without fitting an OU model to observed prices.

For trading, they use tabular Q-learning. The state encodes recent spread moves by direction and magnitude; available actions buy, hold or sell a unit, subject to the current position. The policy enters a long spread and later closes it, rather than initiating a short spread. Its reward favors buying below and selling above the equilibrium mean, with a transaction-cost term $c|A|$. Q-values combine immediate reward with the estimated value of later decisions.

Because the reward requires a true equilibrium mean, the agent trains on 10,000 one-year simulated OU paths before application to historical stocks. The market study forms ten US pairs in 2022 and trades them in 2023. Published daily Sharpe ratios exceed the Distance Method and OU benchmarks for all ten pairs. Those daily figures are not directly comparable with our annualized options Sharpe ratios.

Criticisms that motivate our design

Three issues limit interpretation. **Short history:** one trading year provides limited evidence across regimes. **Costs and scale:** the reward defines a cost term, but its numerical implementation and reconciliation to net performance are not clearly documented. This is not evidence that costs were omitted. Higher turnover raises total costs; larger or faster orders can also increase per-unit slippage and market impact. A constant coefficient does not establish scalability. **Simulation transfer:** formation is empirical, but policy training still uses specified mean-reverting dynamics and a known equilibrium. Real relationships can shift, so extensive synthetic experience does not establish market transfer.

What we borrow and what we change

We borrow the separation of opportunity construction from trading decisions, and the question of whether learning improves on a simple rule. We change the instrument from equity-price spreads to option-implied-volatility spreads, using actual contracts, initial vega matching and daily stock delta hedges. Completed historical trade outcomes replace a known synthetic equilibrium as the basis for our main sizing learner. We make simulated cash flows and costs explicit and extend the historical comparison across multiple trading years.

The next section turns this adaptation into a trading procedure: identify a relative-IV opportunity, buy and sell specific option contracts, and hedge their stock exposure until expiry. We then compare a constant entry size with a size learned from completed historical trades. Finally, we test whether Q-learning improves allocation by accounting for the capital that a trade leaves available for future opportunities. This sequence separates the value of the trading rule from the additional value of learning how to allocate to it.

3 Our strategy and evaluation

3.1 From a volatility signal to a hedged trade

Implied volatility (IV) is the volatility input that reproduces an option quote under a pricing model. Realized volatility (RV) measures subsequent stock-price variation. High IV alone does not establish mispricing: an expensive option may compensate for large future moves. Our proposal combines observable relative IV with relatedness and completed-profitability screens.

We use out-of-the-money puts near 25 delta with approximately 30 days to expiry. Earlier comparisons using the same entry and sizing architecture gave stronger net results for 25-delta puts than for ATM calls or 75-delta puts. A separate comparison of 30- and 90-day contracts on matched proposals also favored the shorter maturity under frozen sizing. These experiments motivated the contract segment used here; their scope is documented in Appendix A. We hold this specification fixed when comparing sizing policies.

For pair A, B, define $D_t = IV_{A,t} - IV_{B,t}$ and rank it within the 252-session rolling history ending on the signal date. Above the 80th percentile, the rule sells A's relatively rich put and buys B's relatively cheap put; at or below the 20th percentile it reverses the direction. Relative richness is measured against the pair's own history, allowing for persistent IV differences. Although the percentile cutoff is fixed, the corresponding IV boundary changes with the rolling sample.

We screen for related stocks using return beta between 0.85 and 1.15 and correlation of at least 0.5. We then check whether historical simulations for the same pair and proposed direction have positive average net profitability. This check uses outcomes completed within the previous 91 and 365 calendar days, requiring at least 21 simulations in the shorter window and 63 in the longer window. If either count is too small or either average is nonpositive, we do not propose the trade. These simulations may overlap and need not have been taken by the funded account.

An outcome becomes available to the screen only after its simulation has finished, strictly before the new signal date. For example, a simulated trade still open on 15 June cannot contribute its eventual expiry profit to a decision made on 15 June. The counts ensure some historical evidence is available; they do not represent 21 or 63 independent observations. The screens complement the IV rank by checking stock relatedness and prior net outcomes for the direction.

Contract quantities approximately match initial vega: the change in option value for a one-percentage-point increase in IV. Table 1 is an illustrative example of the initial positions, with each put at delta -0.25 per share and each contract representing 100 shares.

Position	Contracts	Vega per IV point	Initial stock hedge
Buy A puts	+100	+\$450	Buy 2,500 shares
Sell B puts	−150	−\$450	Short 3,750 shares

Table 1: Initial vega matching. A contracts have \$4.50 vega each; B contracts have \$3.00.

If A's IV rises by two points and B's by one, the local vega approximation gives $\$900 - \$450 = \$450$ before other effects. This is an interim mark change. In the backtest, we hold each option leg to expiry and adjust its stock delta hedge daily, unwinding the hedge after settlement. Option quantities stay fixed after entry, except for documented corporate-event exits. Profit includes premiums, settlement payoffs, stock-hedge gains and losses, dividends, interest and costs.

3.2 Data, execution and a shared account

Historical option quotes, stocks, rates and dividends come through WRDS. Of 345 candidate pairs, usable contracts narrow the prepared panel to 125 pairs, 161 assets and 36,327 dated candidate observations. Screening produces 608 proposals over 2018–2025, including 312 in the later period. Availability therefore narrows the original research universe. The pair-universe appendix lists all 125 pairs in the prepared historical panel, their recorded candidate dates, and the number of screened proposals for each pair. It also distinguishes this panel from the original 345-pair roster and the 46 pairs that generate qualified proposals during 2018–2025.

On the signal date, we select the specific put on each stock, including its strike and expiry. We attempt to trade those same contracts on the next trading day, rather than choosing different contracts after seeing the next day's prices. We round to whole contracts and require the initial long and short vega exposures to differ by no more than 10%. A proposal is rejected if its quotes or account checks fail. All accepted trades use one $10 million account. Stock hedges in the same name are combined into a single net position, and new entries must fit within the capital and risk limits already used by existing holdings.

We book option trades at quoted midpoint and deduct a cash execution penalty equal to 0.10 IV percentage point times the traded vega. For day t, the evaluation costs are

$$C_t^{\text{execution}} = \sum_{j \in \mathcal{F}_t} |\Delta N_{j,t}| \big(0.10\, V_{j,t} + 0.03\big) + 0.0001 \sum_s P_{s,t} |\Delta h_{s,t}|, \tag{1}$$

$$C_t^{\text{funding}} = \Big[0.003\, H_{t-1}^{-} + 0.01 \max(-B_{t-1}, 0)\Big]\, \Delta\tau_t.$$

Here $\mathcal{F}_t$ contains paid option fills, ΔN is the number of contracts traded, and V is one contract's dollar vega per IV percentage point (100 times the per-share quote vega). Thus 0.03 is the commission in dollars per contract. $P|\Delta h|$ is stock-hedge dollar turnover after netting, charged at one basis point. H^- is the previous day's short-stock notional, B is cash, and $\Delta\tau$ is elapsed calendar days divided by 365. The funding rates are 0.3% annual short-stock borrow and a 1% surcharge on negative cash.

Benchmark-rate cash interest is recorded separately as $I_t = r_{t-1} B_{t-1} \Delta\tau_t$; negative cash therefore pays the benchmark rate as well as the surcharge above. Expiry settlement carries no option commission or execution penalty. Archived training paths use $0.65 instead of $0.03 per contract; the effect and remaining limitation are discussed below. Entry admission limits stress exposure, reserves and stock hedges (Appendix A).

3.3 Training, selection and testing

Each month, we train a new tree and size map using historical examples from the preceding 18 months, then apply that model to the next month's entry decisions. Figure 1 separates this rolling process from the choice of model specification. Training and application use successive slices of the same market history.

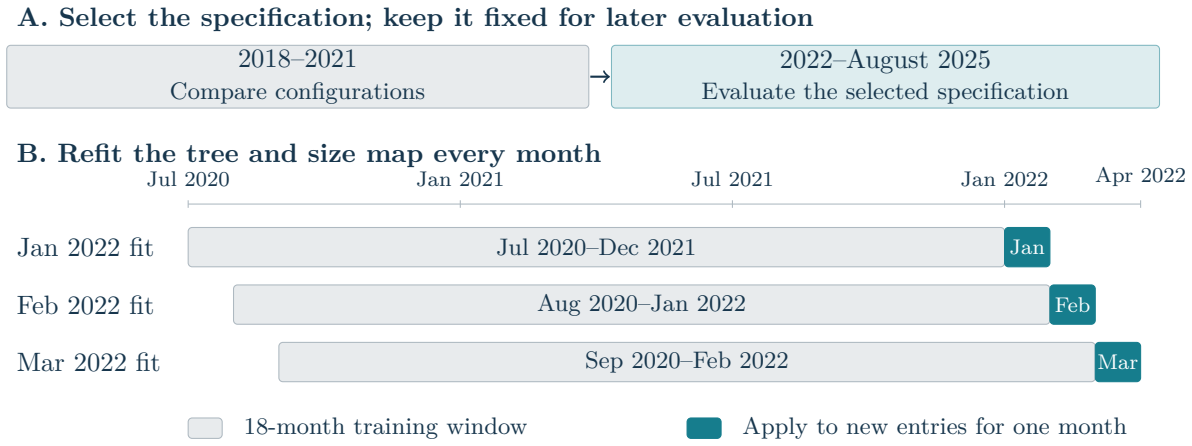


Figure 1: Two decisions, one historical dataset. The upper panel selects the 27-input, 18-month, 50/75/100% specification using earlier performance. The lower panel illustrates monthly refitting during evaluation. A training example is included only if its signal date falls within the window and its trade outcome is completed before that month's fit.

Configuration selection compares 33 saved sizing configurations on 2018–2021 performance. The selected specification stays fixed for the later comparison, while its fitted tree and size map continue to update monthly. Historical examples before 2018 support the initial fit.

The later test covers 3 January 2022–29 August 2025. Configuration choices stay fixed, but the monthly fitting procedure continues. For example, a trade completed in March 2022 may inform an April 2022 fit; a trade that finishes after the April cutoff cannot. This is a rolling train-then-test procedure, not fitting a model using its future test outcomes. Signal generation stops on 30 June 2025, with existing positions followed through the August cutoff. The account carries its existing holdings and cash across the 2022 boundary.

There is a separate research-design limitation: 2022–2025 results had already been examined in earlier strategy development before this split was imposed. The monthly fits respect time order, but the test is not independent of all earlier choices of contracts, screens and universe. We interpret the later comparison as historical evidence rather than independent validation. Selection-account and cost differences are documented in Appendix A; they are not additional fitting stages.

Performance is measured above a cash benchmark because interest materially contributes to account growth. For daily excess returns e_t, annualized excess Sharpe is $\widehat{S} = \sqrt{252}\,\bar{e}/s_e$, with sample standard deviation s_e. Profit above cash, admitted entries and reserves complement Sharpe; drawdown is relative to cash. These distinguish trading performance from interest income and low capital deployment.

3.4 Learning an entry size: decision tree and CEM

The opportunity rule fixes the pair, direction and contracts. We learn only the entry size: 50%, 75% or 100% of a reference budget of $450 vega per IV point on each leg. Opportunities differ in their historical outcomes, so the learner groups them and assigns a common size to each group (Figure 2).

Each monthly regression tree receives 27 inputs: 15 numerical characteristics and 12 availability flags, covering relative IV, IV/RV, relatedness, maturity, VIX, costs and completed profitability. It learns splits using historical full-size profit above cash as its target. Each of its up to eight leaves is a state. The cross-entropy method (CEM) then chooses one size for each state.

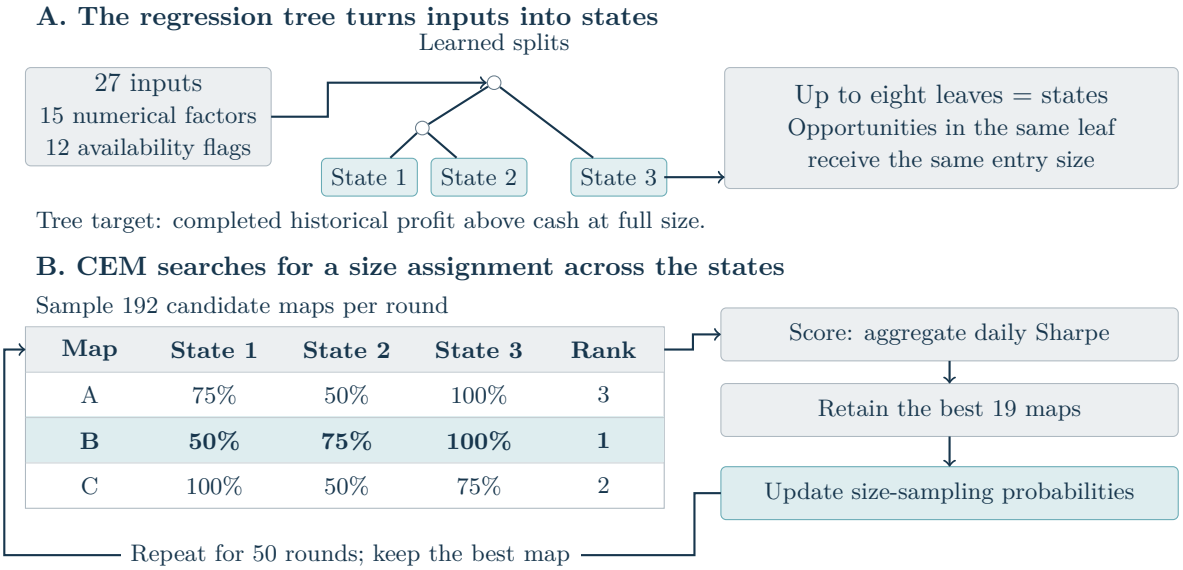


Figure 2: From market inputs to entry sizes. The three-state tree, candidate assignments and ranks are illustrative; actual monthly trees have up to eight states. CEM searches over complete maps, with one 50/75/100% size per state. The selected map is used for the next month's new opportunities.

CEM scores each map using the combined daily profit paths of its historical examples. For leaf $\ell(x_i)$ of example i and its saved daily net profit above cash $p_{i,d}(a)$ at size a, the score is

$$g_d(\boldsymbol{a}) = \sum_i p_{i,d}(a_{\ell(x_i)}), \qquad \widehat{S}(\boldsymbol{a}) = \sqrt{252}\,\frac{\overline{g(\boldsymbol{a})}}{s_{g(\boldsymbol{a})}}, \qquad a_\ell \in \{0.50, 0.75, 1.00\}.$$

Daily paths preserve overlapping trades and zero-profit sessions. Constant-size maps are included as anchors; the search does not establish a global optimum.

For a new qualified proposal, the fitted tree assigns a leaf and its size map supplies the requested entry budget. Account checks then accept or reject it. The tree groups opportunities by full-size profit; CEM chooses sizes by aggregate daily Sharpe. Training sums standalone paths, while funded evaluation applies joint capital admission (Section 4.2).

4 Benchmark and learned-sizing results

4.1 Does learned entry sizing improve the benchmark?

We compare the tree/CEM policy with constant requests of 50%, 75% and 100% of the same reference budget. Each policy receives the same proposals and keeps the proposed direction and contracts.

Figure 3 shows when profit above cash accumulates. Most of the learner's final lead develops during 2022; both policies lose ground during 2024.

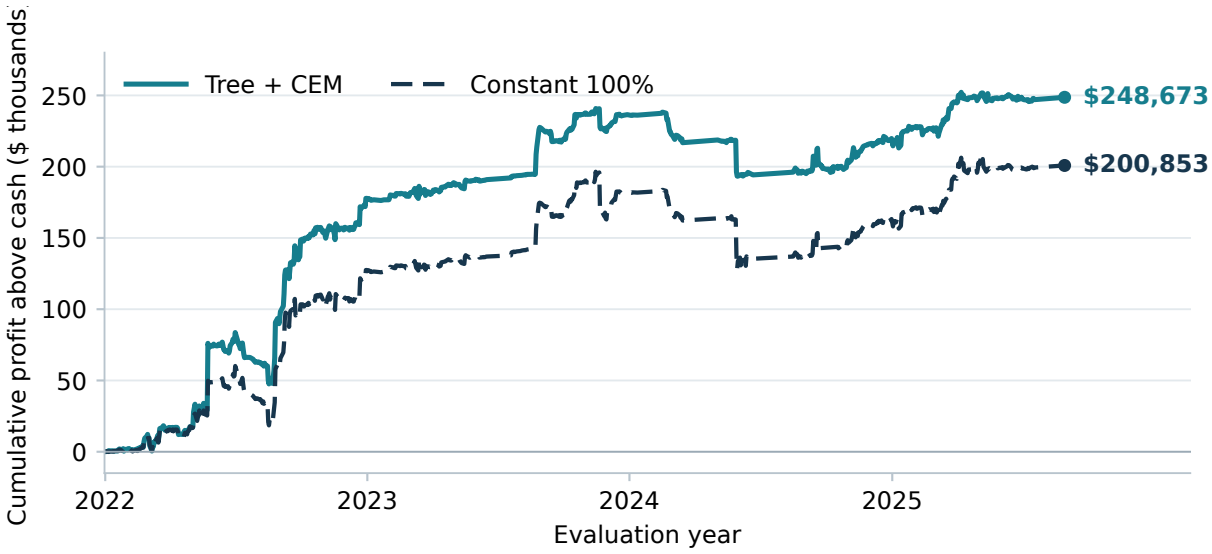


Figure 3: Cumulative profit above cash, 3 January 2022–29 August 2025. Curves are rebased at the 31 December 2021 close; the books retain carried positions. Endpoints match Table 2. The paired 95% interval for the Sharpe gain, $[-0.103, 0.424]$, includes zero.

Table 2 compares the learner and constant sizing with identical proposals and final funded-account rules. Constant full sizing provides a useful historical benchmark: later excess Sharpe is 1.055 and profit above cash is $200,853. Learned sizing raises those estimates to 1.263 and $248,673, an observed Sharpe gain of 0.208 and approximately $47,800 additional profit above cash.

Policy	Earlier Sharpe	Later Sharpe	Later entries	Profit above cash ($)	Relative drawdown (%)
Constant 50%	0.437	1.007	151	204,913	0.735
Constant 75%	0.449	0.944	104	195,021	0.748
Constant 100%	0.478	1.055	75	200,853	0.646
Tree + CEM	0.573	1.263	110	248,673	0.495

Table 2: Earlier: 2018–2021 replay at $10 million; later: January 2022–August 2025. Entries, profit and maximum cash-relative drawdown refer to the later period. Earlier replay scores differ from archived large-account selection scores.

The main saved paired circular-block bootstrap, using 126-session blocks and 1,000 replications, gives a 95% interval of $[-0.103, 0.424]$ for the Sharpe gain. It includes zero and omits search adjustment. The observed gain therefore does not establish a repeatable sizing improvement. The learner admits 110 trades against 75: it changes portfolio access as well as exposure to individual trades.

The account earns annualized total return of 4.667%, while annualized growth relative to an all-cash account is 0.530%. Most total account growth therefore comes from cash interest. Trading is active—110 new entries—but its net contribution above cash is modest: $248,673 over the later period. Average reserve is 2.44% of equity and peak reserve 15.77%. We use excess Sharpe and profit above cash to assess the strategy's added value, rather than attributing the full 4.667% account return to trading.

4.2 Why portfolio admission changes the comparison

A frozen-policy diagnostic makes the allocation mechanism concrete. On all later standalone paths at current costs, an additive sizing alternative has Sharpe 0.726 versus 0.606 for tree/CEM. In funded portfolios, the ranking reverses: 1.013 versus 1.263. Restricting paths to each policy's actual admissions locates that reversal at entry selection under capacity constraints.

Changing commission from the archived training convention improves both standalone scores but does not explain the reversal. Both the additive standalone advantage and funded disadvantage have paired intervals spanning zero. The diagnostic identifies an evaluation mechanism without proving the tree superior: sizing affects the set of accepted trades, so good isolated-trade scores need not produce the best funded book.

Removing a pair or changing an allocation therefore requires rerunning admission, financing and hedges. Subtracting previously recorded profit cannot show how the portfolio would actually have traded. Even low average reserve can coexist with binding constraints within a particular pair or wing.

4.3 How broad is the historical result?

The learner's $248,673 later profit above cash reconciles to $211,262 attributed to pairs and $37,411 of shared financing relative to cash. Financing includes interest on previously accumulated profit; it is not an independent pair signal. Among pairs with nonzero later attribution, 21 are profitable and 11 lose money. The top three positive contributors supply 51.6% of positive pair profit and 66.9% of net pair profit. The largest, SNAP/UBER, has four new entries but only two overlapping-entry episodes.

In 2022, 66 of the learner's 110 later entries generate 83.1% of net later pair profit. Table 3 shows that 2024 has a positive account return but loses relative to cash. Shared stocks and common shocks create dependence across pairs, so trade counts overstate independent market experiences. The aggregate result does not establish a stable benefit across names and regimes.

4.4 Does adapting the opportunity threshold help?

To distinguish allocation from opportunity selection, a supporting test changes only the IV-rank cutoff and keeps the other screens and full sizing. Earlier selection chooses a tighter 5%/95% rule, which later raises Sharpe to 1.290 but admits only 25 trades and earns $161,611 above cash, less

Year	Account return (%)	Above cash (pp)	Excess Sharpe	Entries
2022	3.470	1.673	2.458	66
2023	5.288	0.432	1.492	19
2024	5.104	−0.283	−0.907	11
2025 (partial)	3.207	0.169	1.092	14

Table 3: Tree/CEM annual funded results; 2025 ends on 29 August. Above-cash figures are return differences in percentage points.

than the benchmark. Its Sharpe-gain interval includes zero. Monthly and VIX-dependent cutoffs also fail to establish improvement (Appendix B). Thus the main finding concerns sizing within the retained proposal rule; we have not established a better adaptive opportunity rule.

5 Does sequential learning add value?

5.1 Method: valuing capital left for future trades

The admission finding motivates a sequential policy: capital committed today can prevent a valuable trade tomorrow. Unlike CEM's size-map search, Q-learning attempts to value this future opportunity cost. Its book state includes current inventory and available capacity as well as market and proposal information. The studies test entry allocation and daily management of held positions, using historical simulated transitions between account states.

For state s, action a, next state s' and immediate reward r, fitted Q critics regress toward the Bellman target

$$y = r + \gamma \max_{a'} \widehat{Q}(s', a').$$

The second term is the estimated value of later decisions. A matched myopic control removes that continuation term while retaining the immediate objective and matched inputs and actions. This comparison tests whether looking ahead adds value, rather than attributing every difference to reinforcement learning. The original study trains on 2018–2019 transitions, selects on 2020–2021, and refits before the 2022 test. Its earlier scores therefore have a different selection period from those in Table 2.

5.2 Finding: continuation learning has not improved the benchmark

Earlier validation selects action-specific capacity Q, with Sharpe 0.974 against benchmark 0.821. In the later period its Sharpe falls to 0.802, below the benchmark's 1.055. Across the nine tested Q candidates, none establishes an improvement at both tested block lengths. Matched myopic comparisons also do not establish a reliable benefit from continuation learning.

We additionally test whether choosing a reward aligned with the evaluation metric changes this conclusion. An exact finite-episode Sharpe reward uses increments that sum to final Sharpe minus initial Sharpe at discount $\gamma = 1$. Its later Sharpe is 0.738, compared with 0.726 for Q trained on return with the same episode state, and 1.093 for the matched myopic return control. Neither objective alignment nor the control's small benchmark advantage establishes improvement; their paired gain intervals include zero. Supporting reward variants and intervals appear in Appendix B.

The interpretation is limited by estimation quality. The episode fit has only 229 distinct market decision dates, and its Bellman-target changes do not demonstrate convergence. Counterfactual simulations reuse the same market dates rather than add independent experience. We have identified a reason sequential allocation could matter, but these learners have not demonstrated that they estimate its value well enough to improve trading.

6 Conclusion and limitations

We compare constant sizing with a tree/CEM policy that learns entry sizes from completed historical trades. Both policies use the same screened options opportunities, hedging rules and funded-account constraints.

The historical comparison gives the following findings. The constant-full rule earns positive historical profit above cash. Learned sizing raises the funded point estimate, with a 0.208 Sharpe gain and approximately $47,800 additional profit, but its gain interval includes zero. Allocation also changes which proposals enter the account, explaining why standalone and funded model rankings can differ. Sequential Q-learning has not established an additional benefit from valuing future opportunities.

These results describe a historical simulation with concentrated profits and previously inspected later dates. Training, selection and evaluation differ in capital, commission and admission conventions, and execution and option-exercise handling are simplified. The observed gain does not establish repeatable improvement, live profitability or trading capacity. Most total account growth is cash interest; the strategy's added value is measured above cash.

A Implementation and supporting diagnostics

A.1 Evidence behind the contract segment

The earlier shared-architecture experiment compares ATM calls, 25-delta puts and 75-delta puts using the same IV-rank rule, screens, tree/CEM structure, cost conventions and funded replay. Its positive-size policy has stronger historical net results for 25-delta puts. That experiment uses five-year training windows and a large reference account, so its Sharpe levels are not the final selected-policy results reported in Table 2.

The matched-maturity experiment holds pair, signal date, direction and sizing fixed where both 30- and 90-day contract selections exist on the signal date. It favors the shorter maturity for P25 in that historical comparison. It transfers frozen sizing rather than independently optimizing each maturity. Together these records explain our focus on 30-day P25; they do not establish that this segment is optimal across all contract types and maturities.

A.2 Risk, execution and accounting conventions

Entry admission uses a 5% stress limit, an 80% reserve limit and gross stock hedges no greater than twice equity. Inherited pair/wing checks use a $1 million reference account, while funded comparisons use $10 million. Reserve equals gross option premium plus half gross netted hedge notional; it is not broker margin. Entry checks do not guarantee protection against future losses or broker liquidation. Black–Scholes IV and Greeks use a European exercise approximation for American equity options. Market impact, early assignment and forced liquidation are not fully modeled; the execution penalty is a simulation assumption rather than a measurement of live fills.

Attribution follows calendar cash flows, including carried inventory. Pair cost allocations are additive conventions; shared financing remains separate. Accounting removal of one pair is not a tradable leave-one-pair-out strategy because it does not recalculate admission, cash or hedges.

A.3 Tree states and cross-entropy search

The 33-configuration selection uses a $345 million research account; the selected method and constant controls are replayed at $10 million for the final comparison. This transfers a configuration across capital levels. Archived training paths use $0.65 commission per contract versus $0.03 in funded evaluation. Neither convention is changed in this report.

The 27 inputs describe spread rank, IV/RV levels, relatedness, expiry, VIX, expected costs, completed profitability and availability. The tree targets cumulative full-size profit above cash, with at most eight leaves. For n completed examples, minimum leaf size is $\max(8, \min(63, \lfloor n/8 \rfloor))$. Fewer than 16 examples trigger full-size fallback; the initial fit has 74 examples. Every available action requests nonzero exposure, although account admission can reject it. Each month has its own tree and action map, with potentially changing inputs and boundaries.

CEM searches assignments using completed daily trade paths, including zero-profit sessions. Each of 50 rounds evaluates 192 assignments; the best 19 update action probabilities with 30% prior weight, 70% elite weight and 3% uniform exploration. Constant policies and the incumbent are included; the random seed is 2026. Training combines standalone paths, while final replay applies joint admission. Selection and deployment therefore have account-size and objective differences.

A.4 Threshold and Q-study details

Monthly threshold selection uses the preceding 18 months and defaults to 20% when no candidate meets minimum history and entry requirements. The selected 5% cutoff is at the grid boundary, so the experiment does not locate a global optimum. A separate contextual search selects the VIX-dependent rule.

For the exact episode objective, let S_t be cumulative sample Sharpe and reward $r_t = S_t - S_{t-1}$. At discount $\gamma = 1$,

$$\sum_{t=1}^{T} r_t = S_T - S_0.$$

Moment state and remaining horizon enter the policy, and training intervals stop before the cutoff. Changing only a regression loss would not provide this objective alignment. After 24 fitted-Q iterations, mean absolute Bellman-target change declines from 0.0414 to only 0.0393; a stable fixed point is not demonstrated. Sparse data and off-policy extrapolation remain concerns, without proving that a larger model or longer training would succeed.

B Supporting threshold and reward comparisons

The threshold study holds other screens and full sizing fixed. Earlier selection chooses 5%/95% from six cutoffs spanning 5% to 30%. Monthly selection has no eligible candidate in 13 of 42 later signal months. The 5% choice sits at the grid boundary; it does not identify a global optimum. All gain intervals include zero at both tested block lengths.

Rule	Later Sharpe	Entries	Above cash ($)	Gain: 95% interval
Original 20%/80%	1.055	75	200,853	Reference
Selected 5%/95%	1.290	25	161,611	$[-0.360, 1.095]$
Monthly cutoff	1.037	83	216,896	$[-0.184, 0.140]$
VIX-dependent	1.124	91	251,678	$[-0.158, 0.280]$

Table 4: Funded threshold comparison with carried inventory. Sharpe-gain intervals use paired 126-session blocks and 2,000 resamples.

The reward comparison below retains the reported running, differential and exact episode Sharpe variants. The running-Sharpe policy has no new entries; its positive later result is runoff of inherited positions. The largest later point estimate among the original Q variants, risk-utility Q at 1.072, was not selected for deployment; its gain interval is $[-0.266, 0.331]$.

Policy	Later Sharpe	Above cash ($)	New entries
Threshold, full sizing	1.055	200,853	75
Q: running Sharpe change	0.355	12,065	0
Q: differential Sharpe	0.907	143,224	81
Q: exact episode Sharpe	0.738	62,412	28
Q: return, matched episode state	0.726	118,415	102
Myopic return, matched episode state	1.093	157,821	76

Table 5: Reward-alignment experiments and matched episode controls. These are descriptive later results, not selection by later performance.

Exact episode-Sharpe Q differs from the benchmark by -0.316, with interval $[-1.249, 0.886]$, and from matched episode-return Q by 0.012, with interval $[-1.103, 1.395]$. The myopic control's 0.038 benchmark advantage has interval $[-0.269, 0.409]$. These are conditional paired 126-session intervals, without accumulated-search adjustment.

C Pair universe and date coverage

The original retained roster contains 345 pairs. The prepared historical candidate panel used by the sizing study contains 125 pairs and 161 assets. During the 2018–2025 signal calendar, 46 pairs pass all screens and generate 608 proposals; the selected funded policy admits 207 entries, including 110 in the later test. A proposal is a qualified request, not an executed trade.

The earlier selection period is 2 January 2018–31 December 2021. The later test is 3 January 2022–29 August 2025, with new signals ending on 30 June 2025 and the remaining positions followed through the test cutoff. Historical candidate examples before 2018 support training and profitability screens.

Table 6 lists every pair in the prepared panel. Candidate dates are the first and last recorded signal-date examples in that panel, including examples that fail entry screens; they are not a claim of continuous quote coverage or a pair's executed trading period. The proposal count refers only to qualified signals on the 2018–June 2025 formation calendar. A zero count means the pair supplies historical candidates but no qualified funded request in that calendar. Labels follow the archived roster.

Sources: `reports/atm_nn_cost_hurdles/matched_roster.csv` (complete retained roster, with the included flag), `data/submission/artifacts/contracts/p25_feature_search/training_rows.parquet` (prepared panel), and `reports/p25_memory_ablation/formation_calendar.csv` (per-pair eligible signal calendar).

Table 6: Prepared pair panel, recorded candidate-date ranges and qualified proposal counts. Dates are YYYY-MM-DD.

Pair	Candidate dates	n	Pair	Candidate dates	n
A/TMO	2010-09-14–2025-06-11	12	HES/XOM	2011-03-01–2024-06-06	27
ACLS/LRCX	2018-01-30–2025-01-30	0	HIG/CB	2011-03-01–2025-01-29	0
ADM/BG	2011-09-06–2024-07-23	0	HIG/TRV	2011-06-13–2023-08-07	0
AEM/GG	2010-08-05–2018-10-24	0	HL/KGC	2010-09-01–2024-11-26	0
AEM/NEM	2010-08-05–2023-04-18	0	HWM/GLW	2011-10-04–2016-04-28	0
AEO/GES	2023-02-02–2023-05-19	0	JPM/C	2015-03-02–2022-04-28	4
AEO/GPS	2014-06-03–2018-02-22	0	JPM/WFC	2011-10-04–2023-01-31	22
AEO/TJX	2010-09-01–2022-09-28	9	KBH/LEN	2010-10-05–2025-01-30	2
AIG/CB	2010-11-04–2025-06-30	0	KBH/TOL	2011-05-05–2025-03-26	22
AIG/HIG	2015-02-02–2024-02-20	0	KGC/NEM	2010-11-02–2025-06-23	0
AIG/TRV	2010-11-04–2023-09-20	0	KKR/APO	2022-04-20–2025-06-23	3
AMD/MU	2019-04-01–2023-04-28	23	KSS/BURL	2020-05-06–2023-11-22	8
AMD/NVDA	2012-03-06–2022-12-30	6	KSU/UNP	2010-11-02–2021-11-23	0
AMTD/ETFC	2016-06-16–2019-11-22	0	LEN/TOL	2010-09-02–2023-04-25	0
ANDV/PSX	2013-05-07–2018-04-18	0	LH/DGX	2011-10-10–2017-04-12	0
ARKK/TSLA	2022-08-05–2022-10-31	0	LHX/NOC	2010-10-08–2020-07-27	1
ATI/STLD	2010-08-11–2021-03-05	0	LNG/KMI	2013-05-07–2025-06-18	0
ATVI/EA	2017-03-02–2021-09-23	0	LNG/OKE	2018-08-15–2022-08-25	0
AU/NEM	2011-11-01–2023-06-21	0	LNG/WMB	2013-07-02–2025-06-17	0
BA/RTX	2010-08-05–2023-02-16	0	LYFT/UBER	2020-05-05–2025-06-23	0
BAC/C	2011-01-04–2024-04-30	68	M/KSS	2010-08-05–2022-01-31	14
BAC/JPM	2010-08-05–2025-06-26	76	MAC/SPG	2010-08-16–2024-12-17	0
BAC/WFC	2010-08-05–2025-06-26	14	MAR/HLT	2015-11-02–2025-06-24	5

Continued on next page

Table 6 continued

Pair	Candidate dates	n	Pair	Candidate dates	n
BAX/MDT	2016-10-11–2024-04-19	0	MET/PRU	2010-08-05–2025-06-23	6
BLK/GS	2013-09-03–2023-03-30	3	MGM/WYNN	2010-10-05–2024-09-25	2
BLK/JEF	2022-12-07–2022-12-29	0	MSFT/META	2020-04-01–2020-07-31	1
BTU/AMR	2022-04-11–2025-05-19	0	MSFT/V	2018-04-03–2023-05-31	2
BTU/CEIX	2022-06-16–2025-06-11	0	MU/NVDA	2011-11-03–2020-05-22	0
BTU/HCC	2022-03-23–2025-06-11	0	NEE/FE	2023-11-15–2025-02-20	0
BZH/DHI	2015-02-06–2025-05-27	0	NOC/RTN	2010-08-17–2019-04-25	3
BZH/KBH	2014-10-15–2025-05-19	0	NVTA/NVCR	2020-05-11–2020-06-23	0
BZH/TOL	2013-11-05–2025-05-27	0	OIH/HAL	2015-06-12–2025-05-27	34
C/WFC	2013-01-04–2025-06-26	32	OIH/SLB	2012-08-07–2025-06-25	32
CAL/SKX	2021-04-08–2022-10-27	0	OIH/XOP	2016-11-09–2025-06-25	0
CAR/HTZ	2020-04-01–2020-04-20	0	OKE/KMI	2015-04-21–2025-06-18	0
CCL/NCLH	2020-01-22–2025-06-30	3	OKE/WMB	2015-02-13–2025-06-17	0
CE/HUN	2022-10-25–2025-04-24	0	PENN/CZR	2020-04-07–2025-06-23	2
CENX/AA	2017-06-22–2025-06-16	0	PENN/WYNN	2021-07-02–2022-09-28	0
CHK/SWN	2011-09-06–2019-06-17	0	PLUG/BE	2021-05-14–2024-03-27	0
CLF/FCX	2010-08-05–2022-08-31	0	RCL/NCLH	2016-07-19–2023-02-16	2
CLF/TECK	2010-08-06–2025-06-17	0	RIG/GLNG	2019-02-01–2022-11-21	0
COR/MCK	2017-12-18–2022-10-24	0	RRC/SLB	2011-09-09–2022-01-20	0
CSX/UNP	2010-08-06–2025-05-19	0	SGMS/IGT	2018-09-04–2020-06-25	0
CVX/XOM	2010-08-05–2025-06-26	3	SNAP/UBER	2022-04-14–2022-11-30	7
CYH/THC	2014-09-05–2016-02-12	0	SPWR/CSIQ	2017-04-20–2024-02-21	0
DHI/KBH	2022-08-01–2023-03-29	1	T/VZ	2012-01-03–2023-10-31	1
DHI/TOL	2010-08-10–2023-04-20	2	TER/QRVO	2020-07-06–2023-02-17	1
DHR/TMO	2022-04-04–2025-06-18	23	THC/HCA	2013-07-05–2025-06-30	14
DUK/SO	2010-10-05–2023-09-21	0	THC/UHS	2015-06-01–2015-10-23	0
ELV/CNC	2020-05-05–2020-07-28	0	TLRY/ACB	2021-04-16–2022-04-05	0
ETSY/Z	2022-09-06–2022-09-29	0	TRGP/KMI	2016-09-06–2025-05-22	0
FCX/TECK	2015-09-08–2025-06-17	0	TRV/CB	2011-11-04–2025-06-12	3
FE/SO	2010-11-03–2024-03-15	0	TXT/UTX	2011-05-03–2023-02-16	0
FNSR/VIAV	2017-09-19–2017-12-15	0	UAL/AAL	2014-09-12–2025-06-30	31
GE/RTX	2010-08-05–2022-02-22	2	UAL/DAL	2010-08-05–2025-06-30	36
GES/TJX	2012-03-06–2023-04-10	0	UHS/HCA	2015-11-16–2025-06-20	0
GG/NEM	2010-08-05–2019-01-17	0	UNH/ELV	2011-07-05–2025-06-30	0
GM/FCAU	2023-02-06–2024-02-21	0	UPST/COIN	2022-03-04–2024-02-29	3
GPS/GES	2022-10-07–2023-08-24	0	URI/AER	2018-03-05–2025-05-27	0
GPS/TJX	2010-09-01–2023-01-19	15	WMB/KMI	2015-06-12–2025-06-17	0
HAL/SLB	2010-08-05–2025-06-25	6	WYNN/HLT	2019-02-08–2022-12-28	0
HD/LOW	2016-02-04–2025-06-25	5	XOP/COP	2010-08-09–2025-06-26	17
HES/CVX	2011-02-01–2025-01-16	0			

D Reproducibility and evidence sources

Numerical claims use the following saved repository records.

- Main sizing and annual results: `reports/p25_validation_split/periods.csv` and `annual.csv`. The main gain interval is in `holdout_intervals.csv`, and selection records are in `selection.json` in the same directory. Figure 3 uses the selected daily account in `data/p25_validation_split_experiment.zip` and the constant-full daily account under `data/submission/artifacts/contracts/p25_capital_sensitivity/replays/10m_fixed100`.
- Contract-segment comparisons: `reports/same_structure_smile/findings.txt` and `reports/p25_maturity_matched/Findings.txt`.
- Threshold comparisons: `reports/p25_threshold_search/Findings.txt` and `funded/sharpe_intervals.csv` within that experiment.
- Funded versus standalone diagnostic: `reports/p25_generalization_bridge/Findings.txt`.
- Pair attribution: `reports/p25_strategy_audit/pairs/Findings.txt`, `reconciliation.csv` and `concentration.csv` in the same directory.
- Sequential learning: `reports/p25_rl_research/Findings.txt` and `reports/p25_episode_q/Findings.txt`, with their interval tables; running and differential Sharpe studies are retained under `reports/p25_sharpe_q`.

The main sizing interval uses 1,000 paired circular-block resamples with 126-session blocks. Supporting diagnostic intervals use 2,000 paired resamples and 126-session blocks, with 63-session sensitivity in saved tables. All condition on fitted policies and omit accumulated-search adjustment. Different experiments have different earlier selection periods and carry-in rules; their earlier scores should not be combined into one leaderboard.

In the Q-study audit, full-history constant-full Sharpe is 0.769 (conditional 126-session 95% interval $[0.054, 1.328]$). Later significance depends on block length; this is historical evidence under that procedure, not independent validation.

Vendor data retain their institutional license restrictions. Repository: https://github.com/kreitzz/rl-atm-vega-pairs.

References

[1] B. Ning and K. Lee (2024). *Advanced Statistical Arbitrage with Reinforcement Learning*. arXiv:2403.12180v1. https://arxiv.org/abs/2403.12180.